



Fig. 8: The main window (for low-resolution screens)

EFY Note

The source codes of this project are included in this month's EFY DVD and are also available for free download on source.efymag.com website.

'Save' dialog box is initialised with 'All files' and 'RTF files' filters, the dialog box pops up and the user is asked to enter the file name to save it.

MSComm1_OnComm(). This function handles the different events raised by the comm1 object. The main event used here is the 'comEvReceive' event. Whenever a new data is available to the comm1 object, this 'comEvReceive' event is raised and recognised by the code snippet in this module and as per user settings, is updated to the Receive box. If the user has checked the 'flash on receive' checkbox in the other setting panel, then it will use timer1 object to flash the receive box, and if user has checked the 'speak on receive' then it passes the received data to the 'speak' object to speak the message. Time logging is also managed here if selected.

WriteINIupdate(). This subroutine is used to write the '.ini' configuration file to save all the settings and COM port number so that during the next start, the settings and last port used can be restored for hassle-free operation.

ReadINIupdate(). This subroutine is used to restore all the previous settings and port numbers to the settings panel on the software.

Timer1_Timer(). This routine manages the flashing of the Receive box on each reception if enabled by the user.

Timer2_Timer(). This routine manages the looping of data to be sent multiple times to the serial port as specified by the user at defined time intervals.

COMAvailable(ByVal iPortNum As Integer) As Boolean. This function is used to check whether a serial port is available or not. The port number is the parameter to be passed (by value) and it returns true if the port is available on the user's computer, else false. This is used to generate the list of available serial ports in combo1 box during the form1 load event.

Testing procedure

Most of the testing procedures are already given under 'Running the application' sub-heading earlier. Here are brief steps to test the program:

1. If you have Visual Basic 6.0 already installed on your computer, compile the project as given below.

Open the project folder from this month's EFY DVD and click on 'serial_port_v2.vbp' file. The project will open in the Visual Basic 6.0 IDE. Compile your project into an executable format. Go to the File menu and click on 'Make Serial Port Monitor.exe'. Before running/compiling the program, ensure that following three executable files are installed on your system which are given in the project's folder—The TTS voices (msttsl.exe), the Microsoft agent (MSagent.exe) and the Microsoft command and control engine (actcnc.exe).

2. If you do not have Visual Basic 6.0 installed on your computer, you can run the Setup file included in this month's EFY DVD. It will automatically install the Serial Port Monitor.exe

in your system.

3. When you run the program, the list of available ports is generated and if there is no serial port available on the computer, it will give you a warning message. Before you open the serial port, you need to configure the settings according to your needs like the baud rate, parity, etc.

4. If a valid serial port is available, simply select the COM port from the top-right list box and hit 'Open' button. Now your program is ready to send or receive the data. For example, you can connect a Bluetooth module to your PC. Start the communication by sending text from your smartphone to PC via Bluetooth. You can see the conversation detail in the Send box and Receive box.

You can send and resend the data in loops. As soon as new data arrives, it is updated to the Receive box. You can clear the contents in both the Send and Receive boxes by clicking Clear buttons. You can enable speak/stop speak options.

5. You can save the whole conversation using the Save feature in the File menu bar and the conversation will be saved in rich text file (RTF) format.

6. In the Tools menu, you can open system's calculator, notepad, clipboard and the volume control.

Using the software is very easy. When you open the application, the previous settings will be loaded automatically into the software. The status bar at the bottom of the application will tell you about your current settings and current date and time. The software is fully tested on Windows XP and found to be working satisfactorily.

There are two versions of the GUI developed for this project: one for the screen resolutions equal to or above 1024×768 pixels (as shown in Fig. 8) and another for the screen resolutions equal to or above 1280×1024 pixels. Both are almost the same and so you can use any of them for testing. ●

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